# The Virtue of Complexity in Simple Economic Models

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# Overview

- Recent work by Bryan Kelly, Semyon Malamud and co-authors argue that there is a "virtue of complexity" in empirical asset pricing, meaning we should use models with many, many factors.
- This (perhaps) runs counter to the standard practice of seeking parsimonious models.
- The theoretical part of their work has generated some controversy.
- We focus on the empirical part and ask: does complexity work because the world is complex or does it work even in simple environments?

# Simple Economic Models

- We simulate data from three theoretical models: Berk-Green-Naik (1999), Kogan-Papanikolaou (2014), Gomes-Schmid (2021)
- All are equilibrium models that generate a panel of firms with varying characteristics and expected returns.
  - Expected returns depend on covariances with a stochastic discount factor
  - Covariances are correlated in the cross-section with characteristics
- We evaluate the complexity method by
  - Determining whether performance improves as complexity increases
  - Comparing complex models to Fama-French, Fama-MacBeth, and the CAPM.

# Didisheim-Ke-Kelly-Malamud Characteristic Factors

- Factors = returns of portfolios whose weights are rank-standardized characteristic values in $[-0.5, 0.5]$
- Example: book-to-market factor would be return of portfolio that is long value stocks (above median bm) and short growth stocks (below median bm).
- "More value" $\Rightarrow$ higher weight. "More growth" $\Rightarrow$ more negative weight.

# Composite Characteristics and Factors

- Start with $M$ characteristics $c_{ik}$. Generate $M$ random numbers $h_k$. Compute two new composite characteristics

$$\cos\left(\sum_{k=1}^{M} h_k c_{ik}\right) \quad \text{and} \quad \sin\left(\sum_{k=1}^{M} h_k c_{ik}\right)$$

- Repeat many times to get thousands of new composite characteristics.
- Define a factor as described before for each of the new composite characteristics

# DKKM Empirical Results

- 130 stock characteristics
- Out-of-sample Sharpe ratios reach 4.0 with high-complexity models (10,000+ factors)
- Fama-French-Carhart 6-factor Sharpe ratio $\approx 1.1$

# Data-Generating Models

# Theoretical Models: Overview

- In all models, risk premia depend on covariances with a stochastic discount factor.
- In all models, the covariances are correlated in the cross-section with firm characteristics.
- Hence, characteristics 'explain' risk premia.
- In all models, we can compute size, book-to-market, profitability, asset growth, and momentum (the FFC6 characteristics).
- We can also compute the true traded stochastic discount factor and the true tangency portfolio.

# Berk, Green, and Naik (1999)

- Firms invest optimally given an exogenous pricing kernel
- Fixed number of firms, each receives take-it-or-leave-it investment opportunities each period
- Projects generate operating cash flows until they randomly die
- Investment depends on project NPV (which varies with beta and interest rates)
- Model generates: book value, market value, net income, stock returns
- Characteristics: size, book-to-market, ROE, asset growth, momentum

# Kogan and Papanikolaou (2014)

- Firms invest optimally given an exogenous pricing kernel
- Two aggregate state variables:
  - Disembodied productivity affecting all capital
  - Productivity of newly installed capital
- Firms acquire projects stochastically at firm-specific rates
- Optimal capital investment choice for each project
- Firm-specific and project-specific productivity processes
- Projects produce cash flows until they randomly expire

# Gomes and Schmid (2021)

- General equilibrium model with heterogeneous firms
- Representative household with Epstein-Zin preferences (endogenous SDF)
- Firms make optimal investment and financing decisions
- Single aggregate productivity process (AR(1)) + firm-specific shocks
- Stochastic investment opportunities with random costs
- Lumpy investment (discrete project adoption)
- Debt: consol bonds paying coupons until random expiration
- Tax benefits of debt; costly equity issuance (pecking order)
- Strategic default when equity value $\leq 0$

# Our Simulations

# Advantage of Simulations

- Can compute true conditional expected returns, risks, and Sharpe ratios
- Can compute true traded SDF
- Can compute

$$(\text{estimated SDF value} - \text{true SDF value})^2$$

  each period - the foundation of the Hansen-Jagannathan distance

# Simulation Design

- 1,000 firms and 920 months in each panel (discard first 200 as burn-in, leaving 60 years)
- 12 independent panels for each data-generating model
- In each panel for each model in each month, compute true conditional SDF and true conditional max Sharpe ratio
- Use calibrations from original papers, except we substitute exogenous lognormal SDF in Gomes-Schmid, calibrated to match market risk premium

# Evaluations of Empirical Methods

- Compare conditional Sharpe ratios of estimated MVE portfolios of factors

  Barillas-Shanken, 2017: in horse race between factor models, assuming test assets include competing factors, model with highest Sharpe ratio wins

- Estimate conditional SDFs implied by the models and compute Hansen-Jagannathan distance to true conditional SDF

  HJ distance is the maximum pricing error over all test assets with unit uncentered second moment

- Look at pricing errors for 25 size and book-to-market sorted portfolios

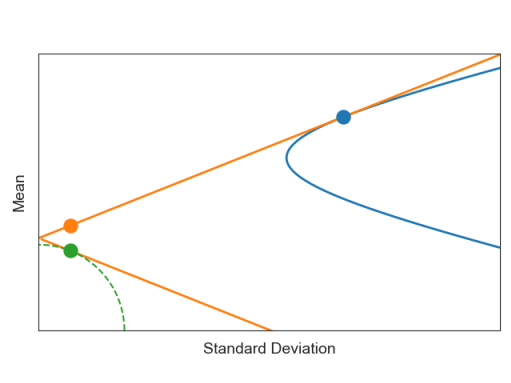
# Britten-Jones Regression

- Can find empirical mean-variance frontier by linear regression of constant 1 on asset excess returns
- Write fitted regression as

$$
\begin{aligned}
1 &= \sum_{i=1}^{N} \hat{\beta}_i (r_i - r_f) + \hat{\varepsilon} \\
&= \hat{z} + \hat{\varepsilon}
\end{aligned}
$$

- Max Sharpe ratio is $\bar{z}/\sigma_z$
- Result originally due to Hansen and Richard for population: projection in population rather than regression in sample

# Britten-Jones Regression




- Green dot $= (1 + r_f)\hat{\varepsilon} - 1$
- Orange dot $= r_f + (1 + r_f)\hat{z}$, the Sharpe ratio of which equals $\bar{z}/\sigma_z$
- Efficient part of frontier is $\{r_f + b\hat{z} \mid b \geq 0\}$

# Our Implementation

- Use known true distributions to calculate $\hat{\varepsilon}$. Use as SDF (orthogonal to excess returns).
- In empirical factor models, use Britten-Jones regression on rolling 360 month windows (following DKKM)
- With many factors (maybe more than 360), use ridge penalization in Britten-Jones regression (following DKKM)

# Performance Measures for Each Factor Model

- Mean theoretical conditional max Sharpe ratio in each panel
- Realized HJ distance: Square root of mean value of $(\hat{\varepsilon}_{\text{factors}} - \hat{\varepsilon}_{\text{all-returns}})^2$ in each panel
- GRS statistic and max Sharpe ratio of unspanned returns of 25 size and book-to-market sorted portfolios

# Ridge Regression

- Minimize: $\frac{1}{T}\sum_{t=1}^{T}(1-\beta' F_t)^2 + \alpha\beta'\beta$
- Penalty parameter $\alpha$ controls shrinkage toward zero
- Essential when number of factors $M$ is large relative to sample size $T$
- We examine performance for various $\alpha$

# Empirical Factors

- Form factors from size, book-to-market, operating profitability, asset growth, and momentum in all models
- Fama-French-Carhart (FFC): usual $2 \times 3$ sorts, use size/book-to-market sort to form SMB
- Fama-MacBeth-Rosenberg (FMR): Fama-MacBeth regressions on characteristics
- Didisheim-Ke-Kelly-Malamud (DKKM): random Fourier features built from the five characteristics plus market return (not penalized in ridge)
- In Berk-Green-Naik, include risk-free rate among DKKM characteristics
- In Gomes-Schmid, include leverage in all methods (for FFC, $2 \times 3$ sort on size and leverage)

# Fama-MacBeth-Rosenberg

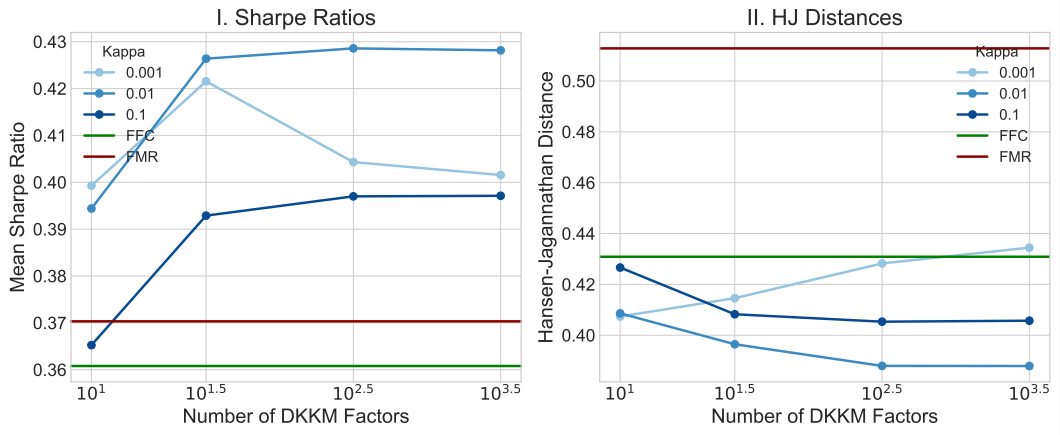
- Fama-MacBeth (1973), Rosenberg (1976), Fama (1976)
- Regression coefficients $(X'X)^{-1}X'R$ are linear combinations of returns $R$
- Set $W = X(X'X)^{-1}$ so regression coefficients are $W'y$
- Regression coefficients are payoffs of portfolios constituting the columns of $W$

# Diversification Property

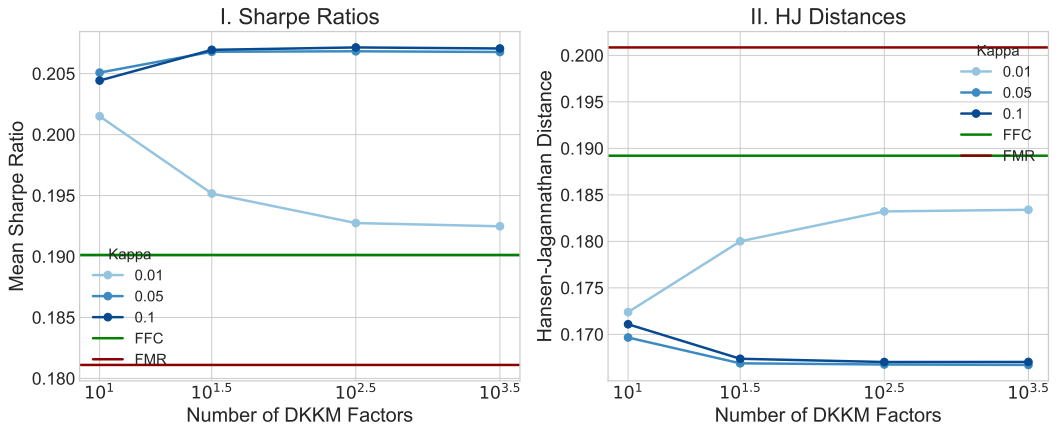
- # columns $W$ = # characteristics $+1$. $W$ satisfies $X'W = I$
- Each portfolio has a unit inner product with the corresponding column of $X$ and is orthogonal to other columns
- Except for the first (intercept) portfolio, each portfolio is a zero-investment portfolio with a unit value for one characteristic and a zero value for all others.
- Many solutions $W$ of $X'W = I$, but projection $W = X(X'X)^{-1}$ solves, for each column, min $w'w$ subject to orthonormal constraint
- We rescale so long and short sides each sum to 1.

# Results

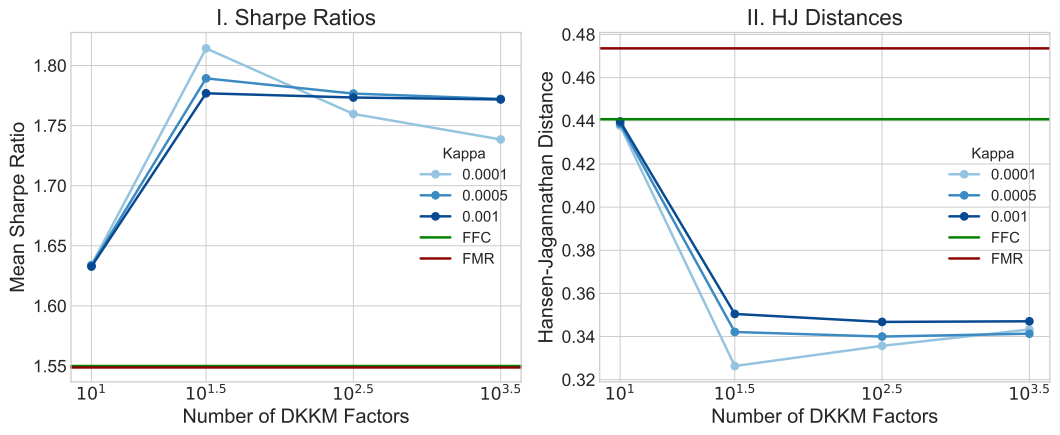
# Berk-Green-Naik Model



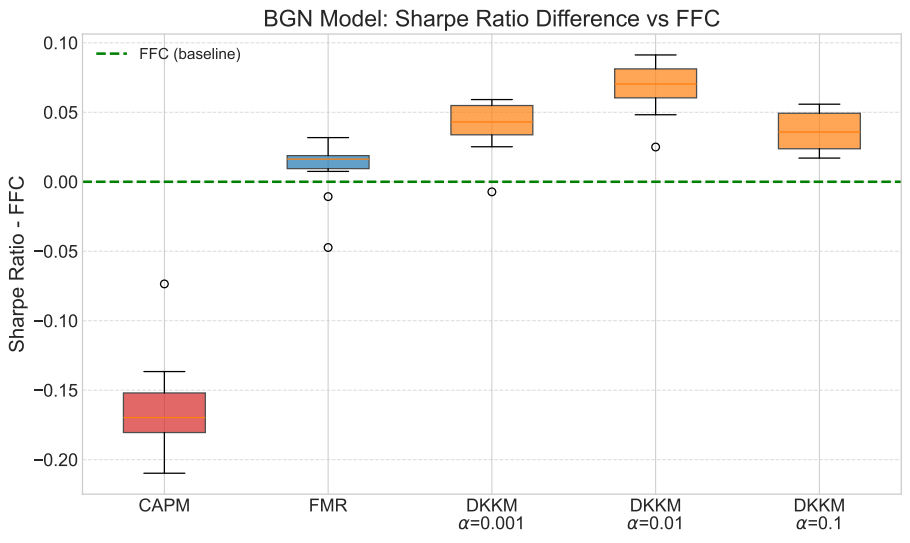
# Kogan-Papanikolaou Model



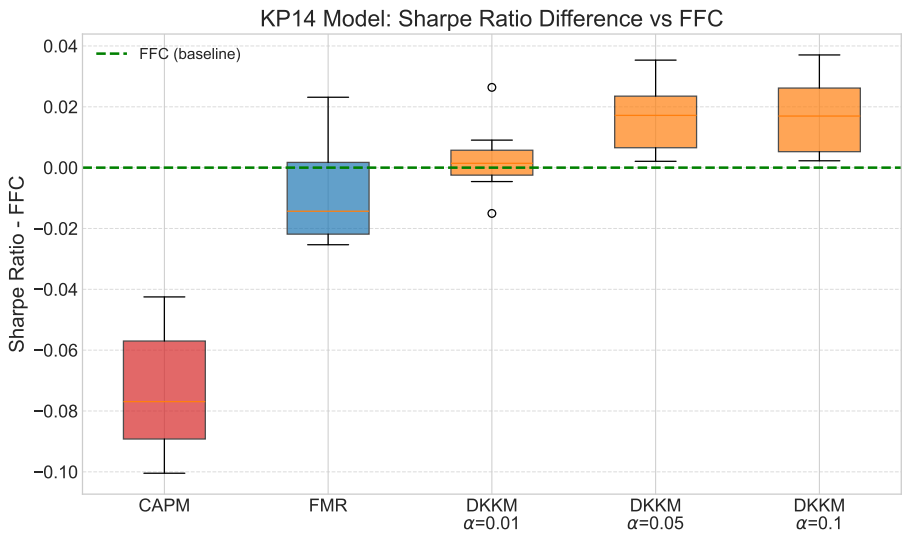
# Gomes-Schmid Model



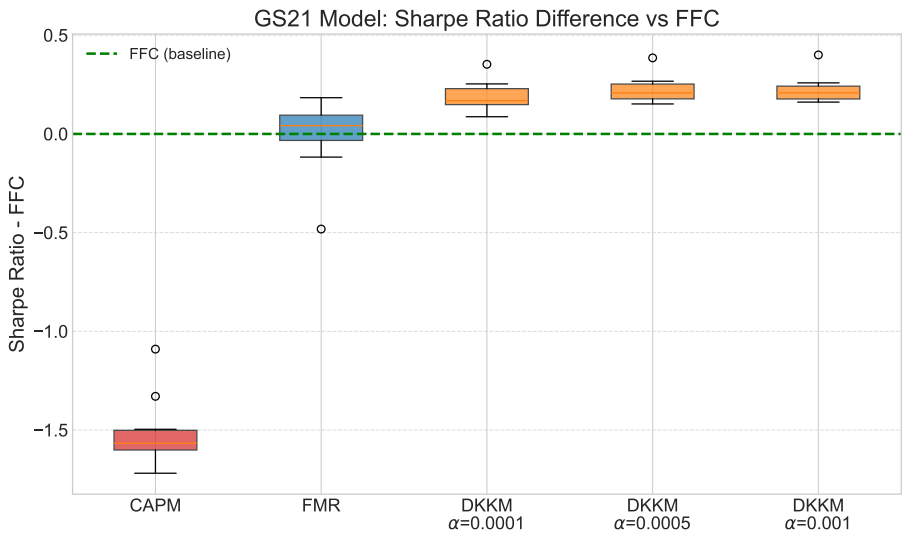
# Sharpe Ratios in Excess of FFC: BGN Model



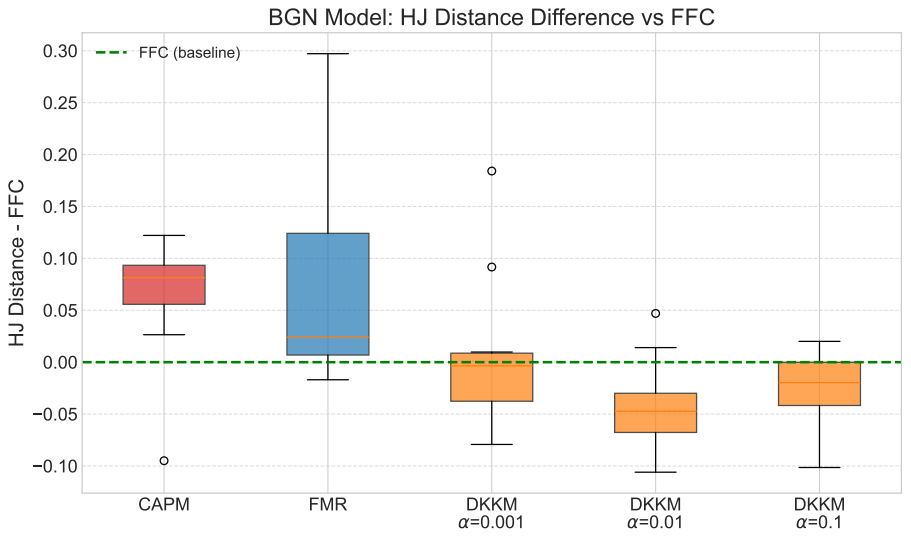
# Sharpe Ratios in Excess of FFC: KP Model



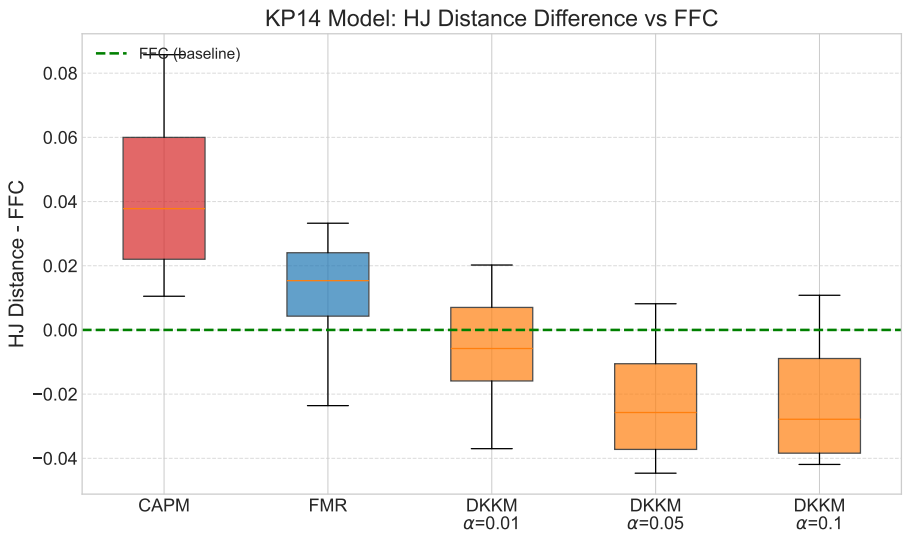
# Sharpe Ratios in Excess of FFC: GS Model



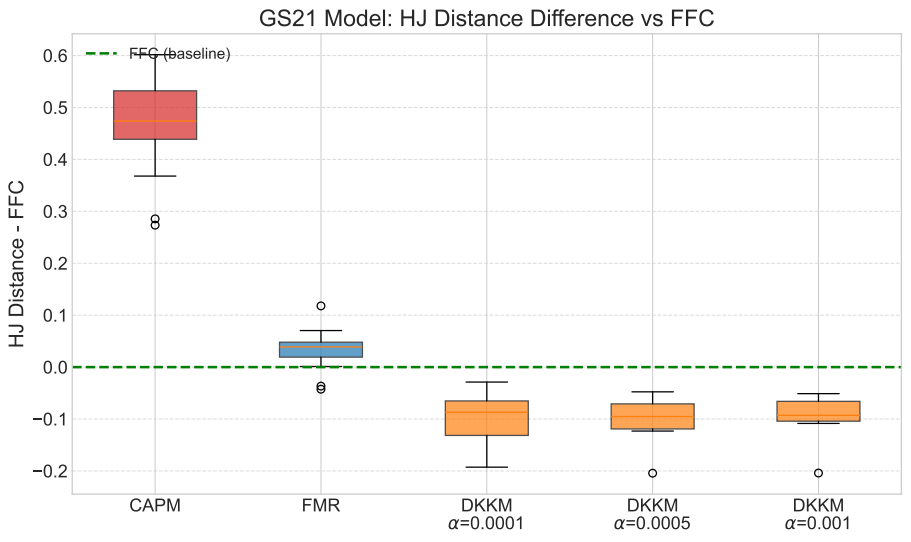
# HJ Distance in Excess of FFC: BGN Model



# HJ Distance in Excess of FFC: KP Model



# HJ Distance in Excess of FFC: GS21



# Statistical Tests: Sharpe Ratios

DKKM (best $\alpha$, 3600 factors) vs other methods

Paired $t$-tests across simulation panels (one-tailed)

| | vs CAPM | | | vs FFC | | | vs FMR | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | Diff | $t$ | $p$ | Diff | $t$ | $p$ | Diff | $t$ | $p$ |
| BGN | 0.232 | 19.5 | $<0.001$ | 0.067 | 12.6 | $<0.001$ | 0.058 | 9.5 | $<0.001$ |
| KP14 | 0.091 | 29.9 | $<0.001$ | 0.017 | 5.0 | $<0.001$ | 0.026 | 4.6 | $<0.001$ |
| GS21 | 1.751 | 45.4 | $<0.001$ | 0.222 | 11.8 | $<0.001$ | 0.223 | 3.4 | 0.003 |

Positive diff = DKKM has higher Sharpe ratio

# Statistical Tests: HJ Distance

DKKM (best $\alpha$, 3600 factors) vs other methods

Paired $t$-tests across simulation panels (one-tailed)

| | vs CAPM | | | vs FFC | | | vs FMR | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | Diff | $t$ | $p$ | Diff | $t$ | $p$ | Diff | $t$ | $p$ |
| BGN | -0.107 | 4.6 | $<0.001$ | -0.043 | 3.4 | 0.003 | -0.125 | 4.2 | $<0.001$ |
| KP14 | -0.064 | 23.7 | $<0.001$ | -0.023 | 4.7 | $<0.001$ | -0.034 | 6.3 | $<0.001$ |
| GS21 | -0.565 | 25.2 | $<0.001$ | -0.099 | 8.3 | $<0.001$ | -0.132 | 6.5 | $<0.001$ |

Negative diff = DKKM has lower HJ distance (better)

# Pricing 25 Size-BM Portfolios

# GRS Test for Pricing Errors

- Regress each of 25 size-BM portfolio returns on factor model return:

$$R_i = \alpha_i + \beta_i F + \varepsilon_i$$

- GRS statistic tests $H_0 : \alpha_1 = \cdots = \alpha_{25} = 0$

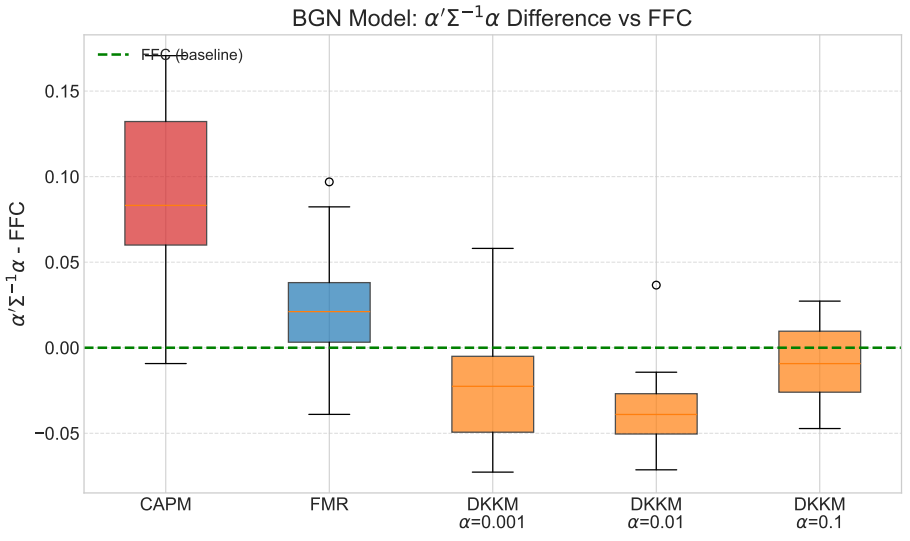
$$\text{GRS} = \frac{T - N - 1}{N} \cdot \frac{1}{1 + \text{SR}_F^2} \cdot \alpha' \Sigma^{-1} \alpha$$

- Key metric:

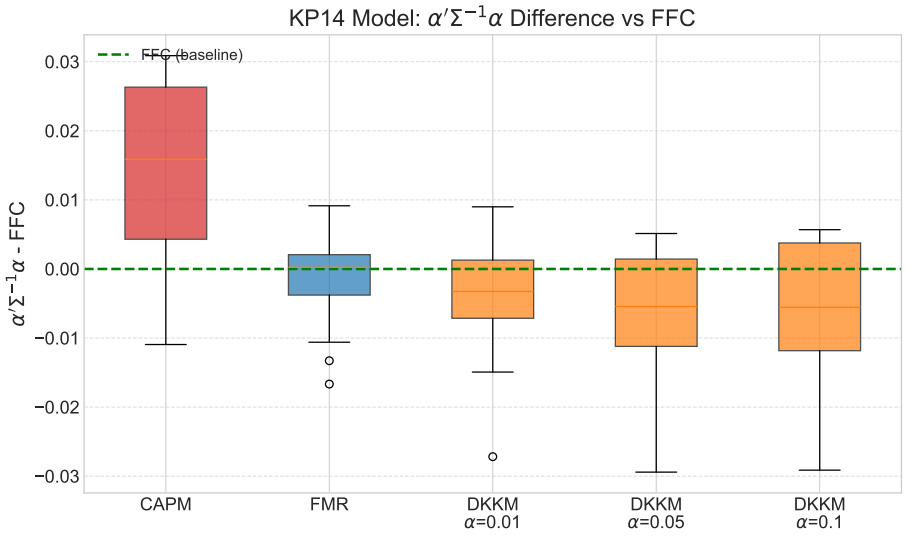
$$\alpha' \Sigma^{-1} \alpha = \text{max Sharpe ratio of} R - F\beta = \alpha + \varepsilon$$

- Under $H_0$: GRS $\sim F(N, T - N - 1)$

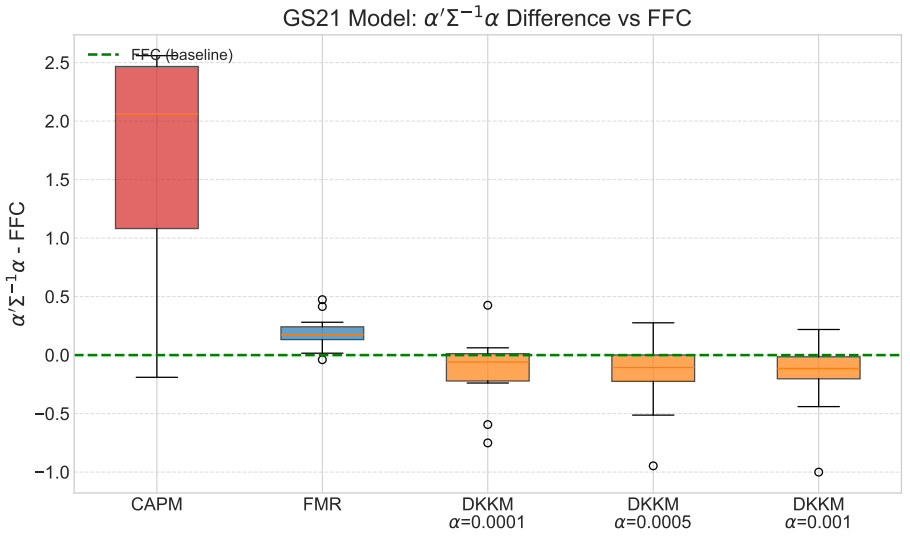
# $\alpha'\Sigma^{-1}\alpha$ in Excess of FFC: BGN Model



# $\alpha'\Sigma^{-1}\alpha$ in Excess of FFC: KP Model



# $\alpha'\Sigma^{-1}\alpha$ in Excess of FFC: GS Model



# Statistical Tests: $\alpha'\Sigma^{-1}\alpha$

DKKM ($\alpha = 0.001$, 3600 factors) vs other methods

Paired $t$-tests across simulation panels

| | FFC | FMR | CAPM |
|---|---|---|---|
| BGN | -0.02 (-2.30)** | -0.05 (-3.49)*** | -0.12 (-8.62)*** |
| KP14 | 0.01 (1.55) | 0.01 (2.39)** | -0.01 (-2.17)* |
| GS21 | -0.17 (-1.91)* | -0.36 (-3.02)** | -1.90 (-7.33)*** |

Format: diff ($t$-stat). Negative diff = DKKM has lower pricing errors (better).
*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

# Statistical Tests: GRS Statistic

DKKM ($\alpha = 0.001$, 3600 factors) vs other methods

Paired $t$-tests across simulation panels

| | FFC | FMR | CAPM |
|---|---|---|---|
| BGN | -0.36 (-2.65)** | -0.66 (-3.71)*** | -1.72 (-11.01)*** |
| KP14 | 0.08 (1.64) | 0.10 (2.36)** | -0.12 (-2.18)* |
| GS21 | -1.55 (-2.61)** | -3.16 (-2.85)** | -30.97 (-12.96)*** |

Format: diff ($t$-stat). Negative diff = DKKM has lower GRS (better).
*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

# Conclusion

- In theoretical economies, we know the true SDF, so can calculate the true errors of pricing models.
- Provides a laboratory for evaluating models
- Results:
  1. More complexity is better
  2. Complexity beats standard models
- This is true even in simple economies!